

Question	Answer	Marks
6(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
6(b)	at least four straight, radial lines to/from surface of sphere	B1
	at least four straight radial lines drawn, approximately equally spaced	B1
	arrows pointing away from the surface of the sphere	B1
6(c)(i)	radius = 3.2 cm	A1
6(c)(ii)	$E = Q / (4\pi\epsilon_0 x^2)$	C1
	$Q = \text{e.g. } 2.2 \times 10^5 \times 4\pi \times 8.85 \times 10^{-12} \times 0.032^2$	C1
	$= 2.5 \times 10^{-8} \text{ C}$	A1
6(c)(iii)	- the (positive) charge is all the way around the surface - a charge placed inside the sphere is pulled equally in all directions - if the field was not zero, the charges would move (until field is zero) - electric field lines go from positive charge to negative charge, and there are no negative charges inside the sphere <i>Any point, 1 mark</i>	B1

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